5. There are a number of factors that should be taken into consideration when deciding what makes the 'best fuel'.

Information was collected about various factors for three fuels, **A**, **B** and **C**.

Fuel A
- Existing supplies will last around 50 years
- Releases 2.8 kJ of energy per gram of fuel burned
- Costs 0.03p per gram of fuel burned
- Burns very easily and no storage issues
- Releases carbon dioxide and water vapour when it burns

Fuel B
- There are infinite supplies of this fuel
- Releases 44.1 kJ of energy per gram of fuel burned
- Costs 0.18p per gram of fuel burned
- Burns very easily but can be difficult to store
- Releases water vapour when it burns

Fuel C
- Existing supplies will last around 250 years
- Releases 1.2 kJ of energy per gram of fuel burned
- Costs 0.04p per gram of fuel burned
- Burns very easily and fairly easy to store
- Releases carbon dioxide, sulfur dioxide and water vapour when it burns

This information was analysed by a group of students to decide what they considered to be the 'best fuel'.

Examiner only

(a) Give the reason why the information about how easily each fuel burns was not useful to the students when reaching their decision. [1]

....

....

(b) One of the students based his decision purely on a judgement of the available supply of each fuel. Choose the order that shows his conclusion. Place a tick (✓) in the appropriate box. [1]

best fuel	**A**	**A**	**B**	**B**	**C**	**C**
	B	**C**	**A**	**C**	**A**	**B**
	C	**B**	**C**	**A**	**B**	**A**
	☐	☐	☐	☐	☐	☐

(c) Which of the following statements best describes how the fuels affect the environment when they burn? Tick (✓) the correct answer and give the reason for your choice. [2]

all of the fuels contribute to acid rain and global warming when they burn ☐

fuels **A** and **C** contribute to acid rain and global warming when they burn ☐

only fuel **C** contributes to acid rain and global warming when it burns ☐

none of the fuels contribute to acid rain and global warming when they burn ☐

Reason ....

....

Examiner only

(d) The cost efficiency of fuel **A** can be calculated as follows:

$$\text{cost efficiency} = \frac{2.8}{0.03} = 93.3\,\text{kJ/p}$$

Use the information given for fuel **B** and this example to calculate the cost efficiency of fuel **B**. [2]

Cost efficiency = ... kJ/p

(e) The students eventually agreed on the following rank order for the fuels.

best fuel

B
A
C

In the table below, tick (✓) **all** the statements that are **correct** and could therefore have been used in deciding upon this order. [2]

	(✓)
fuel **C** will run out after fuels **A** and **B**	
fuel **C** is easier to store than fuel **A**	
fuel **A** burns more easily than fuel **C**	
fuel **B** is the cleanest fuel	
fuel **B** is easier to store than fuel **C**	
fuel **B** will never run out	
fuel **A** is less harmful to the environment than fuel **C**	
fuel **A** is less cost efficient than fuel **B**	

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